

ALVIN EKELUND

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EDUCATION

HARVARD UNIVERSITY

M.S. in Data Science, Institute for Applied Computational Science

Cambridge, Massachusetts

Aug 2026 - Feb 2028

AALTO UNIVERSITY

B.Sc. in Industrial Engineering and Management; Minor in Computer Science

Espoo, Finland

Aug 2023 - Dec 2025

- GPA 4.9/5.0 (Honors); completed B.Sc. in 2.5 years. Thesis on the productivity impact of LLMs in enterprises and startups, graded 5/5; exchange at National University of Singapore.

PROFESSIONAL EXPERIENCE

MIRACLE CONSULTING GROUP

Software Engineer Intern

Helsinki / Singapore / U.S.

May 2026 - Present

- Improving the accuracy and latency of production Graph-RAG pipelines for a Fortune 500 client on Oracle-based enterprise systems.
- Redesigning and developing end-user AI and analytics applications and a unified design system across the client's data products.
- Building an internal agent harness that automates workflows across ERP, CRM, HR and finance systems.

CADENTIA TECHNOLOGIES

Founder's Associate

Helsinki, Finland

Aug 2025 - May 2026

Business Development Intern

Feb 2025 - Aug 2025

- Prototyped knowledge-graph AI features and built sales demos that helped secure the company's first Fortune 500 customer.
- Worked directly with the CEO on research into AI knowledge factories, product iteration and customer-facing demos.
- Automated market and competitive research and built an in-house AI video-production pipeline.

AALTO UNIVERSITY, DEPARTMENT OF COMPUTER SCIENCE

Teaching Assistant

Espoo, Finland

Sep 2024 - Dec 2024

- Taught and helped develop an introductory programming course, mentoring first-year students in debugging and problem-solving.

Additional experience: Apple - B2B and institutional market research; HPP Attorneys - legal AI onboarding strategy (Jan - May 2025).

SELECTED PROJECTS

PERSONAL BRAIN

2026

- Built a local-first knowledge graph and MCP memory server with LLM ingestion, semantic search and Ebbinghaus-style decay (Python, SQLite, Gemini, embeddings).

GROKING INTERPRETABILITY

2026

- Trained a 100K-parameter transformer on modular arithmetic and used Fourier analysis to identify representations behind delayed generalization (PyTorch, Python).

WHAT PREDICTS AI AUTOMATION?

2025 - 2026

- Analyzed 425K Anthropic Economic Index records across 3,259 occupational tasks, finding a 25 percentage-point automation gap between API and chat deployments (Python, Jupyter, O*NET).

WALKTHROUGH - JUNCTION 2025

Nov 2025

- Built a real-time AI grocery assistant for Snap Spectacles; placed Top 5 overall at Junction and 2nd in the Snap Inc. AR Challenge (Lens Studio, Google Cloud, Go).

SKILLS & PERSONAL

Tech: Python, TypeScript, JavaScript, Swift, Scala, SQL, R; PyTorch, React, Oracle APEX, Neo4j/Cypher, Git

Languages: English (Native), Swedish (Native), Finnish (Full Working Proficiency)

Awards: Hack & Roll Singapore main prize (2026); Junction Top 5 and Snap AR 2nd place (2025); Dean's Scholarship (2024, 2025)

Leadership & Other: Aalto Triathlon Club CFO; Slush Team Lead; Finnish Marines NCO course valedictorian (#1/200); Ironman 70.3 Tallinn top national age-group finisher